QueerCase Problem Statements 2025

1. **Anti-Androgens for Transgender Patients**

Anti-androgens are a class of medications that block or suppress the effects of androgens, like testosterone. Existing anti-androgen therapies are primarily designed for cisgender prostate cancer patients and are not optimized for transgender individuals undergoing gender-affirming hormone therapy. These drugs often carry poorly characterized or rare, but serious, side effects, ranging from cardiovascular issues to liver toxicity. There is a pressing need to reimagine anti-androgen strategies that are safer, more effective, and specifically tailored for the needs of transgender patients.

Vasaitis TS, Njar VCO. Novel, Potent Anti-Androgens of Therapeutic Potential: Recent Advances and Promising Developments. Future Medicinal Chemistry. 2010/04/01 2010;2(4):667-680. https://doi.org/10.4155/fmc.10.14

2. **Sustainable Hormone Manufacturing**

Hormone production for medical use, such as estrogen, testosterone, and progesterone, relies heavily on industrial chemical synthesis, which is energy-intensive and environmentally harmful. Emerging research suggests that microbial metabolic engineering could provide greener alternatives, using engineered bacteria to produce hormones at scale. Propose a biomanufacturing strategy using engineered microbes to produce clinically relevant hormones sustainably, with a focus on environmental impact, efficiency, and scalability.

Zhao M, Li X, Xiong L, Liu K, Liu Y, Xue Z, Han R. Green Manufacturing of Steroids via *Mycolicbacteria*: Current Status and Development Trends. *Fermentation*. 2023; 9(10):890. https://doi.org/10.3390/fermentation9100890

3. **Biocompatible Materials for Post-Surgical Scar Healing**

Gender-affirming surgeries and other reconstructive procedures often leave patients with significant scarring, which can affect physical comfort and psychological well-being. Current wound dressings and implants are not always optimized for biocompatibility, controlled healing, or scar minimization. Develop a biocompatible material system to promote optimal wound healing, reduce scar formation, and improve patient recovery following surgery.

Downer M, Berry CE, Parker JB, Kameni L, Griffin M. Current Biomaterials for Wound Healing. *Bioengineering (Basel)*. 2023;10(12):1378. Published 2023 Nov 30. https://doi.org/10.3390/bioengineering10121378

4. **Urethral Tissue Manufacturing**

As part of phalloplasty for transmasculine patients, it is necessary to lengthen the urinary tract while making a neophallus. Current methods use a synthetic section of catheter to connect the neophallus and the bladder. This can lead to higher rates of complications while healing. Tissue engineering with the patient's own cells is a promising method of increasing biocompatibility and reducing complications. Develop a tissue-engineered solution that is compatible and lower in rejection and complication rates for urinal tract lengthening.

de Kemp V, de Graaf P, Fledderus JO, Ruud Bosch JLH, de Kort LMO. Tissue Engineering for Human Urethral Reconstruction: Systematic Review of Recent Literature. PLOS ONE. 2015;10(2):e0118653. https://doi.org/10.1371/journal.pone.0118653

5. **Gene Therapy Delivery for HIV Resistance**

The CCR5-Δ32 mutation naturally protects against HIV infection by altering the CCR5 receptor used by the virus for cell entry. A gene therapy that introduces this mutation, or functionally mimics its effects, could significantly reduce susceptibility to HIV. However, a major barrier lies in designing safe, effective delivery vehicles for this therapeutic approach. Current platforms (e.g. AAV, lentiviral vectors, lipid nanoparticles) each face trade-offs in immune response, precision, and durability. Design a gene therapy delivery system capable of safely and efficiently delivering CCR5-targeted therapies to human cells, with a focus on scalability and long-term efficacy.

Ni J, Wang D, Wang S. The CCR5-Delta32 genetic polymorphism and HIV-1 infection susceptibility: a meta-analysis. Open Medicine. 2018;13(1):467-474. https://doi.org/10.1515/med-2018-0062

6. **Gene Therapy For Broadly Neutralizing Antibody**

Broadly Neutralizing Antibodies are naturally produced in some patients infected with HIV, and target the HIV envelope glycoprotein (env). A gene therapy using AAV viral vectors allowed rhesus macaques to produce env antibodies, resulting in 4 out of 8 monkeys having undetectable simian immunodeficiency virus for over a year. In humans, it is also shown that broadly neutralizing HIV-1 antibodies delay the onset of AIDS of those who produce it by many years. Develop a gene therapy that allows a patient to produce broadly neutralizing antibodies.

Klenchin VA, Clark NM, Keles NK, et al. Adeno-associated viral delivery of Env-specific antibodies prevents SIV rebound after discontinuing antiretroviral therapy. *Sci Immunol*. 2025;10(104):eadq4973. https://doi.org/10.1101/2024.05.30.593694

Papers on Gene Therapy

Pan X, Veroniaina H, Su N, et al. Applications and developments of gene therapy drug delivery systems for genetic diseases. *Asian J Pharm Sci*. 2021;16(6):687-703. doi:10.1016/j.ajps.2021.05.003

Uddin F, Rudin CM, Sen T. CRISPR Gene Therapy: Applications, Limitations, and Implications for the Future. *Front Oncol*. 2020;10:1387. Published 2020 Aug 7. https://doi.org/10.3389/fonc.2020.01387

Paper on CRISPR

Adli M. The CRISPR tool kit for genome editing and beyond. Nature Communications. 2018/05/15 2018;9(1):1911. https://doi.org/10.1038/s41467-018-04252-2

Paper on Induced pluripotent stem cells (iPSCs)

Cerneckis, J., Cai, H. & Shi, Y. Induced pluripotent stem cells (iPSCs): molecular mechanisms of induction and applications. *Sig Transduct Target Ther* **9**, 112 (2024). https://doi.org/10.1038/s41392-024-01809-0